THE DIVISORS OF THE TERMS OF ROW n OF OEIS A176206 ARE, WITH MULTIPLICITY, THE PARTS OF ALL PARTITIONS OF n

MATH0

Abstract. Row n of the OEIS triangle A176206 lists $n-k$ repeated $p(k)$ times for $k = 0, \dots, n-1$, where p is the partition function, and in November 2020 Omar E. Pol conjectured in this entry that all divisors of all terms of row n are also all parts of all partitions of n. We prove the conjecture in its intended form, as an identity of multisets: for every $n \geq 1$, the divisors of the terms of row n, counted with multiplicity, are the parts of all partitions of n, counted with multiplicity; the proof compares the multiplicity of each d on both sides through the recursion obtained by removing one part equal to d. The result is not new in substance: it is equivalent to the classical formula $\sum_{j \geq 1} p(n - jd)$ for the number of parts equal to d in all partitions of n, recorded in OEIS A066633 in 2002 and proved by Ghosh Dastidar and Sen Gupta in 2010, and to a 2023 theorem of Merca, so our contribution is the first formal proof of a known result. The proof is verified in Lean 4 with Mathlib against the frozen, fidelity-reviewed statement that math0 formalized from the OEIS entry, whose definition of the triangle is checked by the kernel against all 94 listed terms.

1. Introduction

Let $p(k)$ be the number of partitions of k (OEIS A000041 [8]), so that $p(0), p(1), p(2), \cdots = 1, 1, 2, 3, 5, 7, 11, \dots$. The OEIS entry A176206 [11] is the irregular triangle whose row $n \geq 1$ is the nonincreasing list R_n in which $n-k$ is repeated $p(k)$ times, for $k = 0, 1, \dots, n-1$:

$$R_1 = (1), \quad R_2 = (2, 1), \quad R_3 = (3, 2, 1, 1), \quad R_4 = (4, 3, 2, 2, 1, 1, 1), \quad \dots$$

Row n has length $\sum_{k<n} p(k)$, which is A000070$(n-1)$ [9]. The entry was submitted by Alford Arnold on April 11, 2010. On November 22, 2020 Omar E. Pol gave it its current name, "row n has length A000070(n-1) and every column k gives the positive integers", and on November 23, 2020 he added the conjecture that "all divisors of all terms in row n are also all parts of all partitions of n".

The worked example in the entry shows what is meant. The divisors of the twelve terms of $R_5 = (5, 4, 3, 3, 2, 2, 2, 1, 1, 1, 1, 1)$ are twelve 1's, four 2's, two 3's, one 4 and one 5, and these are exactly the parts of the seven partitions 5, $4+1$, $3+2$, $3+1+1$, $2+2+1$, $2+1+1+1$, $1+1+1+1+1$ of 5. So the conjecture compares multiplicities. For a positive integer t let $\mathrm{Div}(t)$ be the set of its positive divisors, and for a partition λ let $\mathrm{parts}(\lambda)$ be the multiset of its parts and $m_d(\lambda)$ the number of parts of λ equal to d. Sums of multisets add multiplicities, and $c \cdot M$ is the sum of c copies of M. We consider the multisets

$$\Delta_n = \sum_{k=0}^{n-1} p(k) \cdot \mathrm{Div}(n-k) \qquad \text{and} \qquad \Pi_n = \sum_{\lambda \vdash n} \mathrm{parts}(\lambda),$$

that is, Δ_n is the multiset of all divisors of all terms of R_n and Π_n is the multiset of all parts of all partitions of n.

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Theorem 1. *For every $n \geq 1$ we have $\Delta_n = \Pi_n$. Equivalently, for all integers $n, d \geq 1$,*

$$\sum_{\substack{0 \leq k < n \\ d \,|\, n-k}} p(k) \;=\; \sum_{\lambda \vdash n} m_d(\lambda). \tag{1}$$

The left side of (1) is the multiplicity of d in Δ_n, since $d \in \mathrm{Div}(n-k)$ if and only if $d \mid n-k$, and the right side is the multiplicity of d in Π_n. Writing $n - k = jd$, the left side of (1) equals $\sum_{1 \leq j \leq n/d} p(n - jd)$.

Prior work and status. Identity (1) is classical. We credit the sources found by the math0 literature referees.

- The OEIS triangle A066633 [10] (Naohiro Nomoto, January 9, 2002) gives the number $T(n,k)$ of parts equal to k in all partitions of n, the right side of (1). Its formula section contains the generating function $\frac{x^k}{1-x^k} \prod_{m \geq 1} \frac{1}{1-x^m}$ of column k (Vladeta Jovovic, January 15, 2002), whose coefficients are the sums $\sum_{j \geq 1} p(n - jk)$, and the formula $T(n,k) = \sum_{j<n,\ j \equiv n \pmod k} p(j)$ (José Luis Arregui, April 5, 2002), which is (1). It also records Pol's recursion $T(n+k,k) = T(n,k) + p(n)$ (2012), noted by George Beck (2019) to be Lemma 1 of [3], and a comment of Pol (February 5, 2021) that $T(n,k)$ is the number of k's among the divisors of the terms of the list whose m-th block consists of $p(n-m)$ copies of m, $1 \leq m \leq n$. This list is row n of A340061 [14], the mirror image of R_n, so the comment states the multiplicity form of Pol's conjecture as a fact.
- Ghosh Dastidar and Sen Gupta [2] showed that adding a part k maps the partitions of n bijectively onto the partitions of $n + k$ that contain k (their Lemma 2). They deduced $Q_k(n+k) = Q_k(n) + p(n)$ for the number $Q_k(n)$ of parts k in all partitions of n (Lemmas 1 and 2), and iterated it to $Q_k(n) = \sum_{j>0} p(n-jk) = \sum_{0 \leq i < n,\ i \equiv n \pmod k} p(i)$ (their Result 2), which is (1). The recursion is Lemma 1 of their follow-up paper [3], which attributes the closed form to A066633, and Hazra [4, Lemma 1.2] derives the closed form by counting the partitions with exactly j parts equal to k. These papers use the formula to generalize the theorems of Stanley and Elder on partitions.
- Merca [6, Theorem 1] proved with generating functions that $\sum_{k=1}^{n} f(k) S_{n,k} = \sum_{k=1}^{n} g(k)\, p(n-k)$ for every arithmetic function f and every $n \geq 1$, where $S_{n,k}$ is the number of k's in all partitions of n and $g(k) = \sum_{d \mid k} f(d)$. The two sides are the sums of f over Π_n and over Δ_n (the value t occurs $p(n-t)$ times in R_n), so Merca's theorem for all f is equivalent to Theorem 1; take for f the indicator function of $\{d\}$.
- In the OEIS, the triangle A338156 [13] (Pol, October 14, 2020), whose row n lists the divisors of the terms of R_n, states "The terms in row n are also all parts of all partitions of n" without calling it a conjecture, and A336811 [12] contains a related unlabelled statement (Pol, June 19, 2021). The conjecture also appears, labelled as such, in A340061 [14] (Pol, December 28, 2020). In A176206 itself (revision #107, June 4, 2022) it is still labelled a conjecture, and the referees found no publication that names it as proved.

Since (1) follows at once from formulas recorded in 2002 and proved in 2010, Theorem 1 is a known result, and we classify our contribution as its *first formal proof*. We know of no earlier formalization. Mathlib [5] contains the bijection of [2] (as `Nat.Partition.partitionWithPartEquiv`), which we use, but not identity (1), and the conjecture is not among the statements of the formal-conjectures repository [1] at the commit `c77fdd4` pinned by math0. Our proof is the argument of [2], arranged as a strong induction that compares the two sides of (1) directly. The Lean proof establishes Theorem 1, as the equality of multisets $\Delta_n = \Pi_n$, for every $n \geq 1$; Remarks 10 and 11 are proved on paper only.

2. Statement

The OEIS conjecture. The COMMENTS section of A176206 contains, in a block headed "From Omar E. Pol, Nov 23 2020", the following sentence, verbatim:

> `Conjecture: all divisors of all terms in row n are also all parts of all partitions of n.`

Conjecture 2 (Pol, 2020)**.** *For every $n \geq 1$, the divisors of all terms of row n of A176206, counted with multiplicity, are the parts of all partitions of n, counted with multiplicity; that is, $\Delta_n = \Pi_n$.*

We read both sides as multisets, as the worked example for $n = 5$ does when it counts "twelve 1's, four 2's, two 3's, one 4 and one 5" on each side. Read as sets, the statement would be trivial: both sides are $\{1, \ldots, n\}$, because every $t \in \{1, \ldots, n\}$ occurs in R_n and is a part of the partition $t + 1 + \cdots + 1$ of n.

Lean statement. The statement is the definition `Math0.Statements.A176206.statement` in the math0 statement file `lean/Math0/Statements/OEIS/A176206.lean`, which was reviewed for fidelity and frozen (SHA-256 `4f45cfdc...ddd5c8e`) before any proof attempt. The definitions it uses and the statement read verbatim (docstrings omitted; namespace `Math0.Statements.A176206`):

```
def p (k : ℕ) : ℕ := Fintype.card (Nat.Partition k)

def rowOf (f : ℕ → ℕ) (n : ℕ) : List ℕ :=
  (List.range n).flatMap fun k => List.replicate (f k) (n - k)

def row (n : ℕ) : List ℕ := rowOf p n

def statement : Prop :=
  ∀ n : ℕ, 1 ≤ n →
    (row n : Multiset ℕ).bind (fun t => t.divisors.val) =
      (Finset.univ : Finset n.Partition).val.bind (fun π => π.parts)
```

In Mathlib, `Nat.Partition n` is the finite type of partitions of n, each given by a multiset `parts` of positive integers with sum n. So `p k` is $p(k)$, and `row n` is R_n: the concatenation over $k < n$ of `p k` copies of $n - k$ (as $k < n$, the subtraction in $\mathbb{N}$ is not truncated and every entry is positive). The cast to `Multiset ℕ` forgets the order, `s.bind f` is the multiset sum $\sum_{t \in s} f(t)$, and `t.divisors.val` is the multiset of positive divisors of t, each once; so the left side is Δ_n. On the right, `Finset.univ.val` is the multiset of all partitions of n, each once, so the right side is Π_n. Thus `statement` is Conjecture 2 for every $n \geq 1$, with no extra hypotheses.

Encoding and check against the OEIS data. The name of the entry defines the triangle by two properties, and the statement file proves both:

```
theorem length_row (n : ℕ) : (row n).length = ∑ k ∈ Finset.range n, p k
theorem row_succ (n : ℕ) : row (n + 1) = (row n).map (· + 1) ++ List.replicate (p n) 1
```

The first gives the row length A000070$(n - 1)$. The second says that row $n + 1$ is row n with 1 added to every entry, followed by $p(n)$ ones; with `row 0 = []` it determines every row, and it says that every column reads $1, 2, 3, \ldots$ downwards. The definition is also literally the comment "Row n contains a decreasing sequence where n-k is repeated A000041(k) times, k = 0..n-1" and Paolo Xausa's Mathematica program in the entry.

The file defines the flattened sequence `a` with the OEIS offset 1 and proves two kernel-checked data theorems: `a_data` states that $a(1), \ldots, a(94)$ are the 94 listed terms (rows 1 to 7 and the

first 19 terms of row 8), and `row_five_divisors` states that the multiset of divisors of the terms of row 5 consists of twelve 1's, four 2's, two 3's, one 4 and one 5, the left side of the worked example. Since `Fintype.card (Nat.Partition k)` does not reduce efficiently in the kernel, these checks use a structurally recursive count $\mathrm{pb}(n,m)$ of the partitions of n into parts at most m, given by $\mathrm{pb}(n,0) = [n=0]$ and $\mathrm{pb}(n,m+1) = \sum_{0\le j\le n/(m+1)} \mathrm{pb}(n-j(m+1),m)$, and the theorem `p_eq_pb`: `p k = pb k k`. Its proof identifies $\mathrm{pb}(n,m)$ with the number of partitions of n into parts at most m, using the bijection `partitionWithPartEquiv` of Lemma 7 below. After `row` is rewritten as `rowOf (fun k => pb k k)`, both data theorems are proved by `decide +kernel`: the kernel evaluates the terms and compares them with the listed values. The right side Π_n is not evaluated in the kernel (enumerating `Nat.Partition n` is too slow), and the proof below does not use these checks.

3. Proof

We follow the declarations of the Lean file (namespace `Math0.Frontier.A176206`). For $d,t \in \mathbb{N}$ let $c(d,t)$ be the multiplicity of d in the multiset of divisors of t (`c d t := t.divisors.val.count d`), and for $d,n \in \mathbb{N}$ let

$$F(d,n) = \sum_{\pi\vdash n} m_d(\pi), \qquad H(d,n) = \sum_{k=0}^{n-1} p(k)\, c(d,n-k).$$

Lemma 3 (`c_eq`)**.** *$c(d,t) = 1$ if $d \mid t$ and $t \neq 0$, and $c(d,t) = 0$ otherwise.*

Proof. By `Nat.mem_divisors`, d lies in the finset of divisors of t exactly when $d \mid t$ and $t \neq 0$, and the multiset underlying a finset has no repeated elements. □

Lemma 4 (`lhs_count`)**.** *For all $d,n \in \mathbb{N}$, the multiplicity of d in Δ_n is $H(d,n)$, and the multiplicity of d in Π_n is $F(d,n)$.*

Proof. The multiplicity of d in $\sum_{t\in s} f(t)$ is $\sum_{t\in s}$(multiplicity of d in $f(t)$) (`Multiset.count_bind`). For Π_n this is $\sum_{\pi\vdash n} m_d(\pi) = F(d,n)$, which in Lean holds by definition. For Δ_n it is the sum of $c(d,t)$ over the entries t of the list R_n. The list R_n is the concatenation of blocks, the k-th block consisting of $p(k)$ copies of $n-k$, which contributes $p(k)\,c(d,n-k)$; summing over $k<n$ gives $H(d,n)$. □

Lemma 5 (`F_zero`, `H_zero`)**.** *$F(0,n) = H(0,n) = 0$ for all n.*

Proof. Parts of partitions are positive, so none equals 0. If $0 \mid t$ then $t = 0$, so $c(0,t) = 0$ for all t by Lemma 3. □

Lemma 6 (`F_lt`, `H_lt`)**.** *If $n < d$, then $F(d,n) = H(d,n) = 0$.*

Proof. Every part of a partition of n is at most the sum n of the parts (`part_le`), so no part equals d. For $k<n$ the integer $n-k$ satisfies $1 \le n-k \le n < d$; if $d \mid n-k$ then $d \le n-k$ (`Nat.le_of_dvd`), which is impossible. Hence $c(d,n-k) = 0$. □

Lemma 7 (`F_ge`)**.** *If $1 \le d \le n$, then $F(d,n) = F(d,n-d) + p(n-d)$.*

Proof. Partitions without a part d contribute 0, so $F(d,n)$ is the sum of $m_d(\pi)$ over the partitions π of n that contain d (`Finset.sum_filter_of_ne`, `Finset.sum_subtype`). Removing one part d is a bijection from these partitions onto the partitions of $n-d$, whose inverse adds a part d: this is Mathlib's `Nat.Partition.partitionWithPartEquiv`, which requires $1 \le d \le n$. Reindexing the sum along the inverse bijection (`Fintype.sum_equiv`) and using $m_d(\sigma \cup \{d\}) = m_d(\sigma) + 1$, where $\sigma \cup \{d\}$ denotes σ with one more part d, we get

$$F(d,n) = \sum_{\sigma\vdash n-d} \big(m_d(\sigma)+1\big) = F(d,n-d) + \#\{\sigma \vdash n-d\} = F(d,n-d) + p(n-d). \qquad \square$$

Lemma 8 (`H_ge`)**.** *If $1 \le d \le n$, then $H(d,n) = H(d,n-d) + p(n-d)$.*

Proof. Write $n = r + d$. Splitting the range $0 \le k < r + d$ into $k < r$ and $k = r + x$ with $0 \le x < d$ (`Finset.sum_range_add`) gives

$$H(d, r+d) = \sum_{k=0}^{r-1} p(k)\, c(d, r+d-k) + \sum_{x=0}^{d-1} p(r+x)\, c(d, d-x).$$

For $k < r$ we have $r+d-k = (r-k)+d$, so $d \mid r+d-k$ if and only if $d \mid r-k$ (`Nat.dvd_add_self_right`), and both numbers are nonzero; by Lemma 3, $c(d, r+d-k) = c(d, r-k)$, and the first sum is $H(d,r)$. In the second sum $1 \le d-x \le d$, and $d \mid d-x$ forces $d \le d-x$, that is, $x = 0$. So only the term $x = 0$ survives (`Finset.sum_eq_single`), and it equals $p(r)\, c(d,d) = p(r)$. Hence $H(d,n) = H(d,r) + p(r)$ with $r = n - d$. □

Proposition 9 (`F_eq_H`)**.** *$F(d,n) = H(d,n)$ for all $d, n \in \mathbb{N}$.*

Proof. For $d = 0$ this is Lemma 5. Let $d \ge 1$; we use strong induction on n. If $n < d$, both sides vanish by Lemma 6. If $n \ge d$, Lemmas 7 and 8 give $F(d,n) = F(d,n-d) + p(n-d)$ and $H(d,n) = H(d,n-d) + p(n-d)$, and $F(d,n-d) = H(d,n-d)$ by the induction hypothesis, since $n - d < n$. □

Proof of Theorem 1 (`result`). Let $n \ge 1$ (the hypothesis is not used). Two multisets are equal if every d has the same multiplicity in both (`Multiset.ext'`). By Lemma 4 and Proposition 9, the multiplicity of d in Δ_n is $H(d,n) = F(d,n)$, which is its multiplicity in Π_n. □

Remark 10 (Closed form)**.** For $d \ge 1$, iterating Lemma 7 and ending with Lemma 6 gives $F(d,n) = \sum_{j=1}^{\lfloor n/d \rfloor} p(n - jd)$, the formula of Arregui in A066633 [10] and of [2, Result 2]. It is not formalized.

Remark 11 (A340061 and Merca's identity)**.** Row n of A340061 [14] lists m repeated $p(n-m)$ times for $m = 1, \dots, n$, that is, the entries of R_n in reverse order, so Theorem 1 also proves the conjecture as recorded there. Summing a function f over both sides of $\Delta_n = \Pi_n$ gives Merca's identity [6, Theorem 1]. Neither deduction is formalized.

4. Machine verification

Files and declarations. The proof is in the file `lean/Math0/Frontier/OEIS/A176206.lean` of the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file `lean/Math0/Statements/OEIS/A176206.lean` (Section 2) and the audit commands in `lean/Math0/Audit.lean`. The main declaration is `Math0.Frontier.A176206.result`, of type `Math0.Statements.A176206.statement`. The auxiliary definitions `c`, `F`, `H` and lemmas are private; each lemma of Section 3 carries the name of its Lean declaration.

The main theorem and the final audit command read verbatim:

```
theorem result : statement := by
  intro n _
  refine Multiset.ext' fun d => ?_
  rw [Multiset.count_bind, Multiset.count_bind, lhs_count, ← F_eq_H]
  rfl

#math0_settles Math0.Frontier.A176206.result Math0.Statements.A176206.statement
```

The file contains no `sorry`, `native_decide`, `set_option` or macros, and, unlike the data checks of the statement file (which it does not use), the proof does not use `decide`.

Audit. The command `#math0_settles`, defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself, fails unless (1) `statement` is a `Prop`-valued definition without universe parameters; (2) `statement` and `result` use no axioms besides `propext`, `Classical.choice` and `Quot.sound`, which rejects `sorryAx`, the axiom `Lean.ofReduceBool` of `native_decide`, and custom axioms; and (3) the type of `result` is syntactically the statement or its negation, up to metadata, with no unfolding. Running `lake env lean Math0/Frontier/OEIS/A176206.lean` in the directory `lean` prints

```
math0 audit: 'Math0.Statements.A176206.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A176206.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A176206.result' proves 'Math0.Statements.A176206.statement'
```

so `result` proves the statement and depends only on the three permitted axioms.

Toolchain and reproduction. We use Lean 4 [7] `v4.33.1` and Mathlib [5] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [1] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh` from the repository root; it fetches the Mathlib build cache, rebuilds all math0 results (re-running every audit command), checks that the pinned dependencies are unmodified, and reports the number of passed statement checks. For this result alone, run the `lake env lean` command above in `lean/` after building its imports with `lake build Math0.Statements.OEIS.A176206 Math0.Audit`.

References

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[9] The OEIS Foundation Inc., `https://oeis.org/A000070` (partial sums of A000041), entry by N. J. A. Sloane.

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[11] The OEIS Foundation Inc., `https://oeis.org/A176206`, entry by A. Arnold, Apr. 11, 2010; name and conjecture by O. E. Pol, Nov. 22–23, 2020; revision #107, Jun. 4, 2022.

[12] The OEIS Foundation Inc., `https://oeis.org/A336811`, entry by O. E. Pol, Nov. 20, 2020.

[13] The OEIS Foundation Inc., `https://oeis.org/A338156`, entry by O. E. Pol, Oct. 14, 2020.

[14] The OEIS Foundation Inc., `https://oeis.org/A340061`, entry by O. E. Pol, Dec. 28, 2020.